SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I G. Amrutha (192511250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Amrutha.

Assessment Tasks

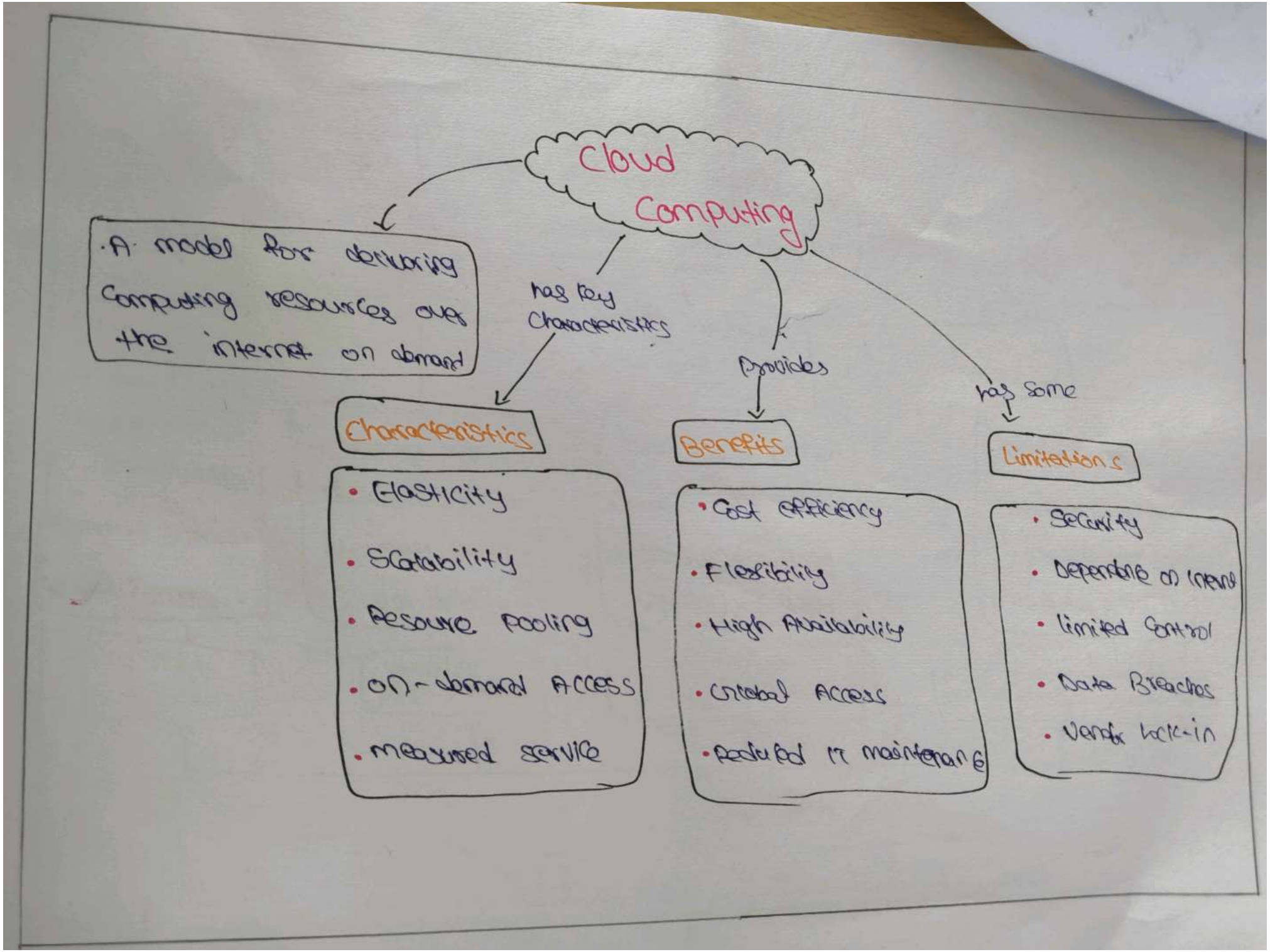
Concept Mapping Tasks

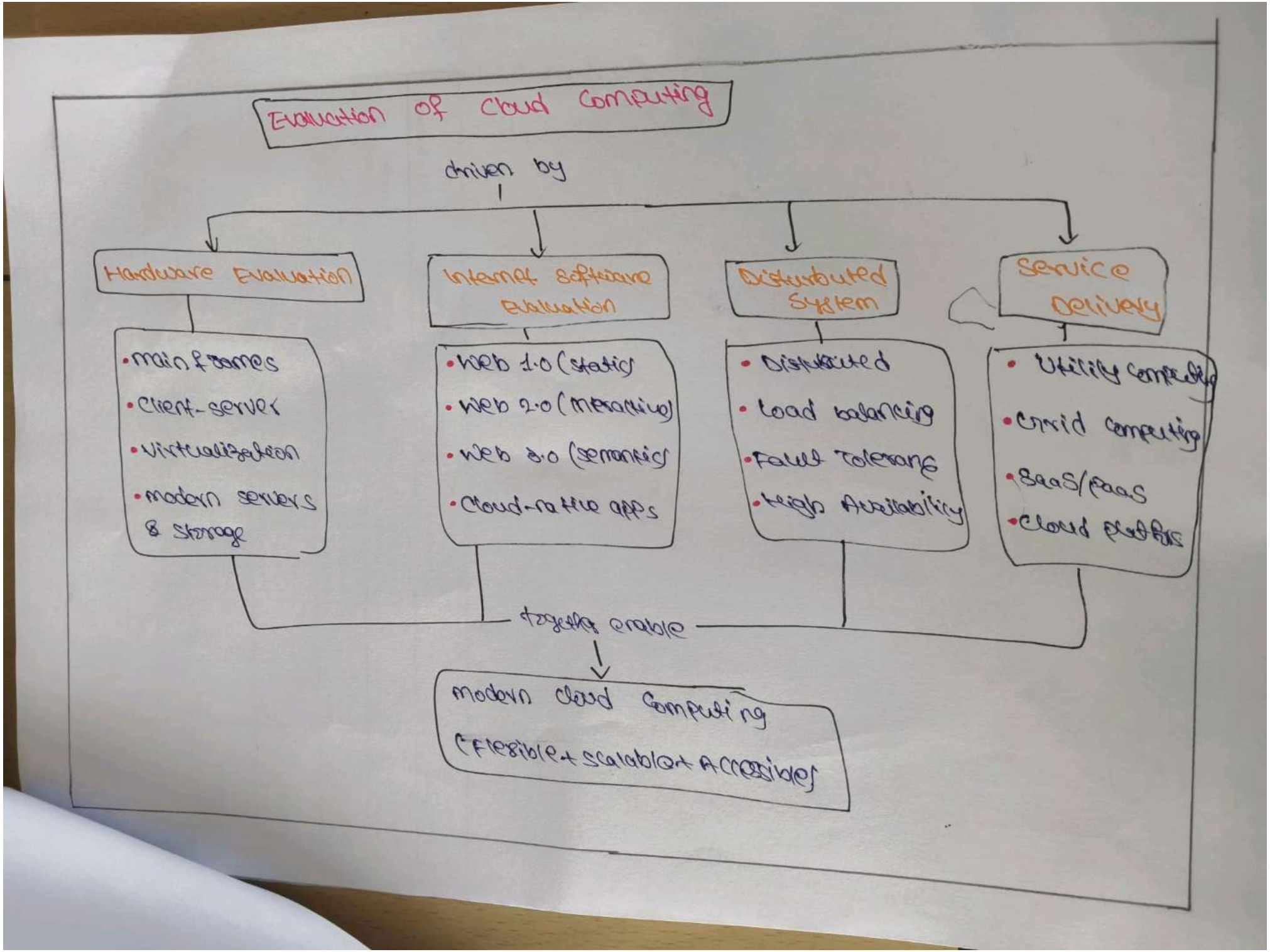
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- *Elasticity, scalability, resource pooling, on-demand access, measured service*
- *Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer*
- *Infrastructure layer, platform layer, application layer, shared responsibility*


Cloud Computing
A model for delivering computing resources over the internet on demand
has key characteristics
Provides
has some
Characteristics
Elasticity
Scalability
Resource pooling
On-demand access
measured service
Benefits
Cost efficiency
Flexibility
High Availability
Global Access
Reduced IT maintenance
Limitations
Security
Dependence on internet
limited control
Data Breaches
Vendor lock-in


Evaluation of Cloud Computing
driven by
Hardware Evaluation
Internet Software Evaluation
Distributed System
Service Delivery
• main frames
• Client-server
• Virtualization
• modern servers & Storage
• web 1.0 (static)
• web 2.0 (interactive)
• web 3.0 (semantic)
• Cloud-native apps
• Distributed
• load balancing
• Fault tolerance
• High Availability
• Utility computing
• Grid computing
• SaaS/PaaS
• Cloud platforms
together enable
modern cloud Computing
(Flexible + Scalable + Accessible)

Cloud Computing:

Cloud Computing is a model for delivering computing resources such as compute power, storage, network and applications over the internet on demand. It has key characteristics like elasticity, scalability, resource pooling, on demand access and measured service. It provides many benefits such as cost efficiency, flexibility, high availability, global access and reduced IT maintenance. However, it also has limitations like security and privacy risks, dependence on the internet, limited control, data breaches and vendor lock-in.

Evaluation of Cloud Computing:

Cloud Computing evolved through four major streams. Hardware evolution provided the computing power. Internet software evolution (web 1.0 to cloud) enabled global access and web-based applications. Distributed systems introduced concepts like load balancing and fault tolerance. Finally service delivery models combined these to deliver modern cloud computing, which is flexible, scalable and accessible.

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented